

Ethan B. Chen

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PROFESSIONAL SUMMARY

Computer Science student at Vanderbilt University building full-stack AI applications and machine-learning systems. Developed technical projects across LLM evaluation, computer vision, business automation, and audio signal processing. Completed an AI-focused internship at Cleveland Clinic and holds certifications spanning machine learning, networking, and hardware.

EDUCATION

Vanderbilt University, School of Engineering

Expected May 2030

B.E. in Computer Science, Minor in Data Science | Nashville, TN

EXPERIENCE

Radiology Department Intern | Cleveland Clinic

Summer 2025

- Worked in an AI-focused radiology environment with exposure to machine learning, image processing, and healthcare AI under Po-Hao Chen, MD, MBA, Vice Chair of AI, Diagnostics Institute.

R&D Intern | Hyland Software

Summer 2024

- Evaluated Hy-Tech Club technical course content and redesigned the program homepage to improve usability and interactivity.

Founder & Web Developer | Wifi Concierge | [GitHub](#)

2024-Present

- Founded a community technology initiative serving 70+ individuals and families; raised \$1,500+ to expand access and built and maintained the public-facing website.

Technology Department Intern | PCs for People

Summer 2025

- Reconditioned and configured computers for affordable resale and supported device preparation for underserved families.

PROJECTS

The Great Prompt-Off | [GitHub](#)

Next.js, React, TypeScript, Supabase/PostgreSQL, OpenRouter

- Built a full-stack AI evaluation platform for a prompt-engineering challenge using synthetic, non-PHI knee MRI reports, with hidden-answer scoring and live leaderboards; mentored by Po-Hao Chen, MD, MBA and Chintan Shah, MD.
- Implemented Supabase-backed persistence, admin controls, LLM evaluation, regression testing, and security boundaries.

TowerLogic | [GitHub](#)

Python, PyTorch, YOLOv8, Computer Vision

- Applied YOLOv8/PyTorch to interpret live Clash Royale gameplay, integrated detection outputs with automated decision logic, and refined behavior through iterative testing.

Piano MIR Performance Assessment | [GitHub](#)

Rust, Python, Audio Signal Processing

- Developed a score-aligned MIR system comparing recorded performances with symbolic score data using note/chroma evidence, timing, and alignment analysis; mentored by Wenbing Zhao, Ph.D., Cleveland State University.

Carat Coin - Kitco Price Scraper | [GitHub](#)

JavaScript, Node.js, Puppeteer

- Built a business automation tool that scrapes Kitco precious-metals bid/ask and high/low data, schedules collection with node-cron, and exports structured CSV data.

SKILLS

Languages: Python, JavaScript, TypeScript, Java, Rust, HTML/CSS | **Frameworks/Platforms:** React, Next.js, Supabase, PostgreSQL, OpenRouter, Vercel | **AI/Data:** PyTorch, YOLOv8, computer vision, machine learning, image processing, audio signal processing | **Tools:** Git/GitHub, REST APIs, Puppeteer, Node.js, Vitest

CERTIFICATIONS

CompTIA A+; CompTIA Network+; HarvardX Machine Learning and AI with Python (CS109xa); LSE Online Ethics of AI; Duke University Java Programming: Solving Problems with Software; IBM Java Programming for Beginners